Higher Physics

Assignment: mark it yourself



A Scottish Borders Council resource, curated by R Stewart

Remember

This pack is for practice **before** you write your report. It is never used while you write it.

Your tick list is the official **Instructions for candidates** in the Higher Physics coursework assessment task. They are the only tick list you may use in the report stage.

Read this on your iPad. Write your answers on the printed **worksheet**.

Contents

Part 1: How to use this pack
Part 2: The marks and the tick lists
Part 3: Script 1: Planck's constant from LEDs
Part 4: Script 2: Internal resistance
Part 5: Script 3: Peak a.c. and d.c. voltage
Part 6: Script 4: Irradiance and distance
Part 7: Script 5: Refractive index of water
Part 8: Answers

1 How to use this pack

Five real assignments are in this pack. They come from Qualifications Scotland's Understanding Standards website. A marker has already marked them.

Your job: mark them yourself first. Then compare your marks with the marker's.

Steps

Step 1: **Read** the marks and tick lists in part 2.

Step 2: **Read** one script: parts 3 to 7.

Step 3: **Fill in** that script's grid on your worksheet.

Step 4: **Complete** the focus tasks on your worksheet.

Step 5: **Check** your marks against the answers in part 8.

Why marking works

The course reports advise this kind of practice. When you mark, you see what the marker looks for. You learn the standard, not someone else's words.

Do not copy sentences from these scripts. Your report must be your own work, about your own experiment.

For teachers

Use the pack during the course and in the research stage. Collect the worksheets in before the report stage.

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2 The marks and the tick lists

The assignment is worth 20 marks. Most marks are lost in the first three sections below. The last column is the summary table from your Instructions for candidates.

Item	Section	Marks	What the marker looks for
8	Evaluation	3	Three evaluative statements supported by justifications.
2	Underlying physics	3	A description of the physics relevant to your aim, which shows your understanding.
5	Uncertainties	2	Scale reading uncertainties shown for all measurements and random uncertainty in measurements calculated.
7	Conclusion	1	A conclusion relating to your aim based on all the data in your report.
4c	Plotting and line	1	All data points plotted accurately and, where appropriate, line or curve of best fit drawn.
6	Analysis	1	Discussion of experimental data.
3e	Citation and reference	1	A citation for an internet/literature source and the reference listed later in the report.
3c	Data table	1	Data from your experiment, including any mean and/or other derived values, presented in a table with headings and units.
3a	Method summary	1	A brief description of an approach used to collect experimental data.
3d	Second data	1	Numerical or graphical data relevant to your experiment obtained from an internet/literature source, or raw data relevant to your aim obtained from your second experiment.
4b	Labels and units	1	Suitable labels and units on the axes.
1	Aim	1	A description of the purpose of your investigation.
3b	Raw data	1	Sufficient raw data from your experiment.
4a	Scales	1	The axes have suitable scales.
9	Title and structure	1	An informative title and a structure that can easily be followed.

Sections ranked by the share of marks lost in the Scottish Borders, 2024 to 2026.

Tick lists from your Instructions for candidates

Underlying physics. You must describe the physics relevant to your aim. You must use your own

words as much as possible. You may choose to include:

- relationships or equations
- definitions of symbols used
- explanations or justifications of relationships or equations
- explanations of physical properties
- copies of diagrams that you would find difficult to draw

Description of experiment(s). Your description must include the measuring instruments you used.

Experimental data.

- Make sure you include column headings. You must also include units, where appropriate.
- You must calculate mean values for your repeated measurements. These must be included in your table.
- Any derived values needed for graphing must be included in your table.

Graph.

- You must produce a scatter graph of your experimental data.
- If you are using graphing software, you must include both major and minor gridlines, and use plotting symbols that are clear but not too large.
- The graph(s) must be large enough to allow points to be read accurately and have suitable scales. It/they must also have labels and units on the axes.

Uncertainties. You must include scale reading uncertainties for all the measurements you have made in your experiment, and calculate the random uncertainty in your repeated measurements.

Citation and reference. One way is to put a number, for example (1), next to the data/information and the same number beside the reference included later in the report. A URL cannot be a citation.

Analysis. You must show a sample calculation. You must state the physical significance of any gradient or constant you calculate.

Conclusion. You must state a conclusion that relates to your aim and is supported by all the data included in your report.

Evaluation. You must make three statements, supported by justification, which evaluate the data/information you have included. You could comment on:

- the accuracy and precision of your experimental measurements
- the adequacy of your repeated readings
- the adequacy of the range over which you altered variables
- the adequacy of how you controlled variables
- any limitations of the equipment you used
- the reliability of your methods
- sources of uncertainty in your measurements

Evaluation

The list is a menu, not a quota. Pick the three items with the **most evidence** in your own results, and give a reason for each. A statement scores only if it fits this experiment.

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3 Script 1: Planck's constant from LEDs

This assignment investigated the value of Planck's constant from the voltage that switches on LEDs. It is 7 pages long.

Read the whole script first. Then **fill in** grid 1 on your worksheet.

Candidate 1 evidence, Physics Higher Assignment 2023-24, Understanding Standards. © Qualifications Scotland (SQA).

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Higher Physics Assignment

Aim: to determine Planck's constant using the voltage required to light different coloured LEDs.

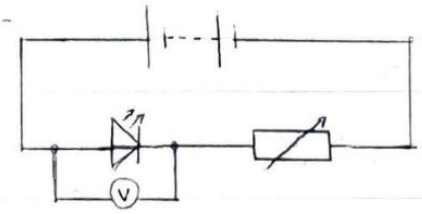
Underlying Physics:

Planck's constant is a highly important number which relates the energy of a light photon to its frequency. It is used in various aspects of physics, such as Schrodinger's Equation and de Broglie's relation for wavelength of matter waves. Planck's constant can be determined using LEDs.(1)

LEDs, when supplied with enough voltage, emit photons. When the minimum energy required to emit a photon, is measured and the wavelength of photons emitted is measured, it can be used to calculate the energy, hf, of the photon.(2) In the equation $E=hf$, h is Planck's constant. Planck had discovered as the energy of radiation increased, so did its frequency, leading to the Planck's constant equation. The equation $h=\frac{E}{f}$ proves that, given that as frequency increases energy increases, h is a constant. Planck's constant can also be determined using $E=h\frac{c}{\lambda}$,(3) using the speed and wavelength of a photon, where $c=3\times10^{8}ms^{-1}$

Experimental Method:

Different coloured LEDs were used in a circuit, and the voltage across them was changed until they first began to emit light. This voltage was recorded for each different LED. The process was repeated so that a mean could be calculated.



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Data:

LED colour	λ of light (nm)	QV	frequency (Hz)	voltage required to light (V) 1	2	3	4	MEAN
violet	405	4.14×10^{-19}	7.41×10^{14}	2.60	2.59	2.58	2.59	2.59
blue	470	3.78×10^{-19}	6.38×10^{14}	2.43	2.32	2.35	2.32	2.36
green	525	3.30×10^{-19}	5.71×10^{14}	2.10	2.03	2.06	2.03	2.06
yellow	590	2.72×10^{-19}	5.08×10^{14}	1.75	1.67	1.66	1.70	1.70
orange	605	2.53×10^{-19}	4.96×10^{14}	1.60	1.61	1.55	1.54	1.58
red	640	2.32×10^{-19}	4.69×10^{14}	1.46	1.45	1.44	1.43	1.45

mean voltage calculation:

$\frac{2.60+2.59+2.58+2.59}{4} = 2.59\text{ V}$

$\frac{2.43+2.32+2.35+2.32}{4} = 2.355 = 2.36\text{ V}$

QV calculation:-

$QV = 1.6\times10^{-19}\times 2.59$

$= 4.144\times10^{-19} = 4.14\times10^{-19}$

$QV = 1.6\times10^{-19}\times 2.36$

$= 3.776\times10^{-19} = 3.78\times10^{-19}$

frequency calculation:

$f = \frac{v}{\lambda}$

$= \frac{3\times10^{8}}{405\times10^{-9}}$

$= 7.407407...\times10^{14}$

$= 7.41\times10^{14}\text{ Hz}$

$f = \frac{v}{\lambda}$

$= \frac{3\times10^{8}}{470\times10^{-9}}$

$= 6.382978...\times10^{14}$

$= 6.38\times10^{14}\text{ Hz}$

Uncertainties

Scale Reading Uncertainty on voltmeter : ±0.01 v

Random Uncertainties:

violet : 0.005

blue : 0.03

green : 0.018

yellow : 0.02

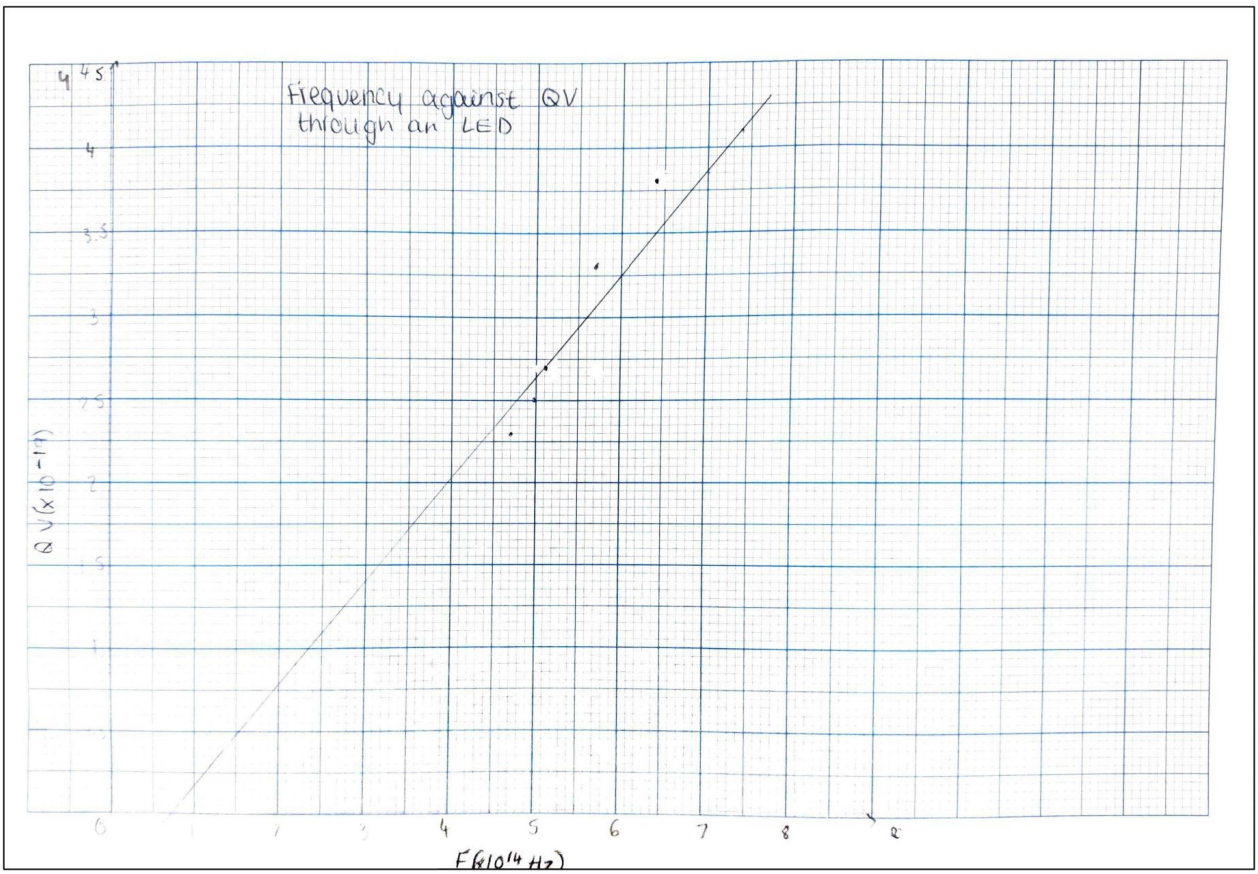
orange : 0.018

red : 0.008

random uncertainty calculation:

$\frac{2.60-2.58}{4} = 0.005$

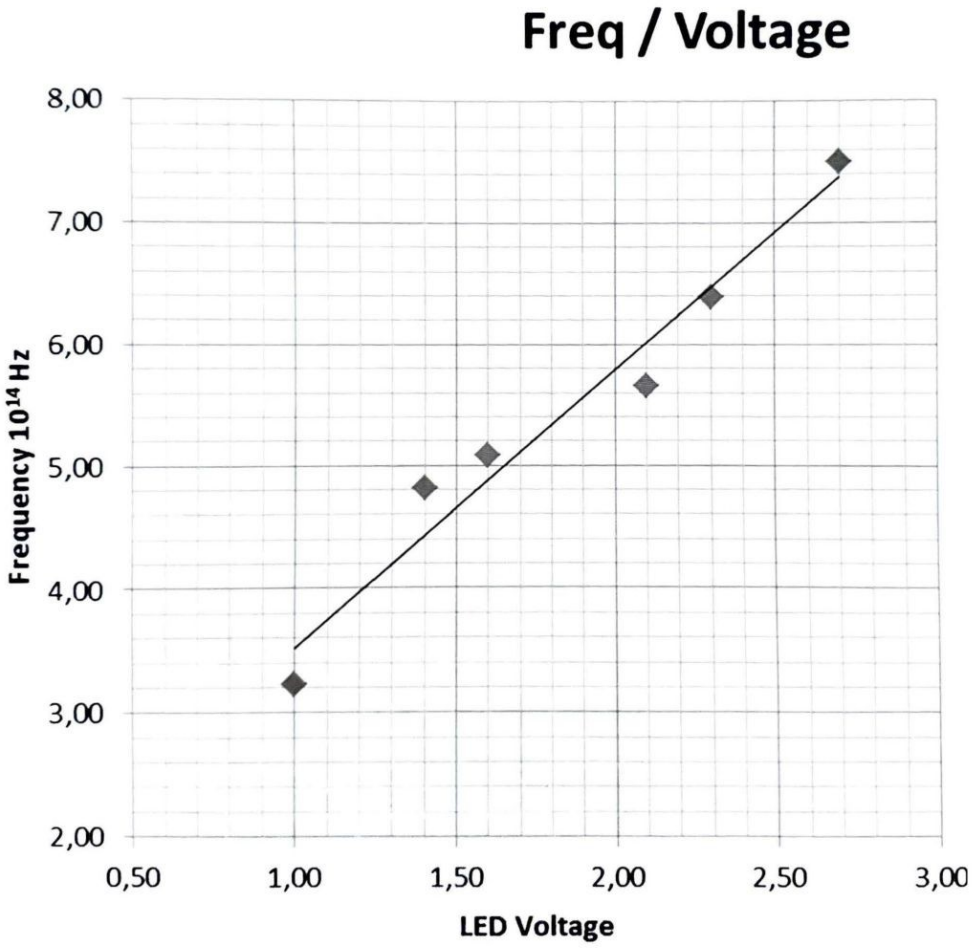
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Data

LED Type	λ (nm)	Freq. (10^{14}Hz)	Voltage (V)
Infrared	930	3.22581	1.0
Red	624	4.80769	1.4
Yellow	590	5.08475	1.6
Green	530	5.66038	2.1
Blue	470	6.38298	2.3
UV	400	7.50000	2.7



http://physicsopenlab.org/2015/12/08/planck-constant-measurement/

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Second source
(graph) (4)

Analysis:
source 1 calculation

$m = \frac{y_2 - y_1}{x_2 - x_1}$

Points: (6, 3.25)
(7, 3.35)

$= \frac{3.35 \times 10^{-19} - 3.25 \times 10^{-19}}{7 \times 10^{14} - 6 \times 10^{14}}$

$= 6 \times 10^{-34}$ Js

source 2 calculation (4)

$m = \frac{y_2 - y_1}{x_2 - x_1}$

Points (2, 5.8)
(2.2, 6.2)

$= 2 \times 10^{14}$

$h = \frac{V}{f} \times Q$

$= \frac{1}{m} \times Q$

$= 8 \times 10^{-34}$ Js

Conclusion:
Planck's constant can be roughly determined by use of LEDs. Given the voltage required to emit photons from various LEDs, Planck's Constant can either be worked out using $h = \frac{QV}{f}$ or by finding the gradient of a graph of QV against f.

Evaluation:
Source 1 was more accurate for determining Planck's constant than source 2. This may relate to the amount of repeats done. Source 1 may be more accurate because more repeats have been made, and a more accurate average taken to be used in calculations.
However, source 1 does not give the same value for Planck's constant as the official value, 6.63×10^{-34}, and the line of best fit on the graph does not go through the origin. This is due to systematic uncertainties. A systematic

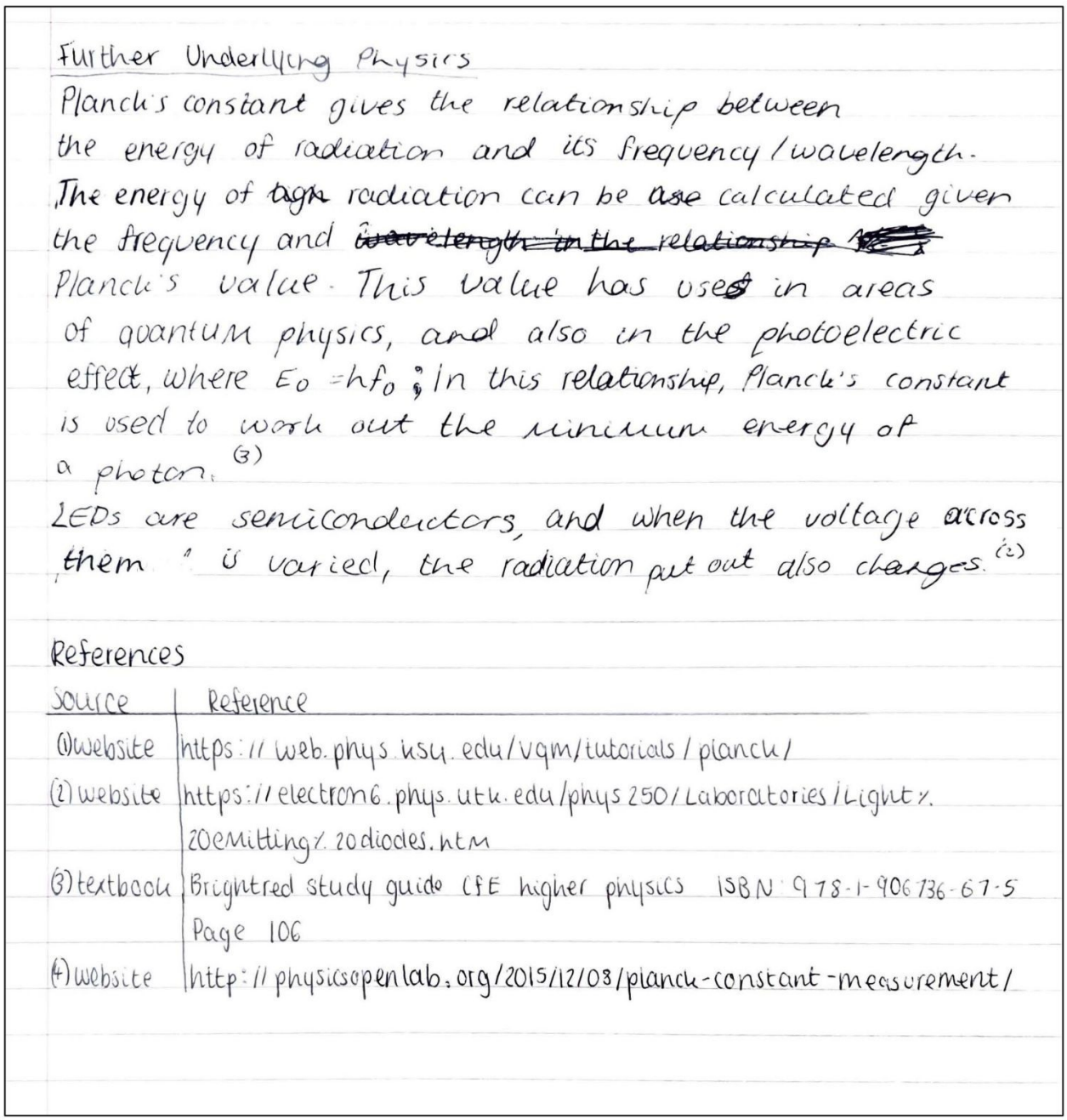
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Uncertainty in this experiment could be an error in the equipment used - the voltmeter's reading, or some of the components of the circuit may have been inaccurate, thus causing the readings taken to be inaccurate.
Another cause of less accurate results could be human error. The experiment
The results could also be less accurate because the experiment was user sensitive. Human error could mean that the switch on voltage of the LED was determined inaccurately - meaning it was not as precise as it could be were technology used to detect when the LED first began to emit light. The ~~experiment~~ results could be more accurate if the experiment was carried out in a darkened room, so that the first emission of light from the LED could be better determined.

References:

Source	Reference
(1) website	https://web.phys.ksu.edu/vqm/tutorials/planck/
(2) website	https://electron6.phys.utk.edu/phys250/Laboratories/Light%20emitting%20diodes.htm
(3) textbook	Brightred study guide CfE higher physics ISBN: 978-1-906736-67-5 Page 106
(4) website	http://physicsopenlab.org/2015/12/08/planck-constant-measurement/

~~Further Underlying Physics~~



Further Underlying Physics

Planck's constant gives the relationship between the energy of radiation and its frequency / wavelength. The energy of ~~light~~ radiation can be ~~use~~ calculated given the frequency and ~~wavelength in the relationship~~ Planck's value. This value has used in areas of quantum physics, and also in the photoelectric effect, where $E_0 = hf_0$; In this relationship, Planck's constant is used to work out the minimum energy of a photon. (3)

LEDs are semiconductors, and when the voltage across them " is varied, the radiation put out also changes. (2)

References

Source	Reference
(1) website	https://web.phys.ksu.edu/vqm/tutorials/planck/
(2) website	https://electron6.phys.utk.edu/phys250/Laboratories/Light%20Emitting%20diodes.htm
(3) textbook	Brightred study guide CfE higher physics ISBN 978-1-906736-67-5 Page 106
(4) website	http://physicsopenlab.org/2015/12/03/planck-constant-measurement/

End of script 1. **Fill in** grid 1 on your worksheet now.

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4 Script 2: Internal resistance

This assignment investigated the internal resistance of a battery, by two methods. It is 9 pages long.

Read the whole script first. Then **fill in** grid 2 on your worksheet.

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Determining Internal Resistance of an Electrical Supply

Aim: To determine the internal resistance of a D battery.

Underlying Physics

All electrical components have resistance. Resistance is the opposition to charge passing through a component. Resistance is measured in Ohms (Ω) & is represented as R in equations.

Electromotive force (emf) is a property of electrical supply components. emf is defined as the ~~Energy~~ ← Energy supplied per coulomb of charge (1) passing through it. To measure emf a voltmeter must be connected across supply, meaning emf is the ~~Pd~~ Potential Difference (p.d.) of an open circuit. emf is measured in Volts (V) and is represented by $\mathcal{E}$ in calculations.

When a current is drawn from supply the voltmeter reading decreases. This is due to ~~the~~ a property of supply components called Internal Resistance. Internal Resistance causes volts from supply to be ~~transforms~~ and lost from the circuit. This decreased voltmeter reading is called terminal potential difference (t.p.d.) This is represented as V_{tpd} and is measured in volts. To find V_{tpd} one can use ~~Ohms Law~~ $V_{tpd} = Ir$ where V_{tpd} is the terminal potential difference, (V) I is the current (Amps, A), and r is the internal resistance of supply (Ω)

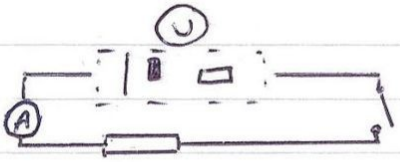
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THE DIFFERENCE BETWEEN EMF & TPD IS 'LOST VOLTS' VOLTS LOST FROM THE CIRCUIT DUE TO INTERNAL RESISTANCE.

THIS GIVES RISE TO THE EQUATION:
- $\epsilon = V_{LOST} + V_{TPD}$ WHERE ϵ IS EMF MEASURED IN VOLTS (V), V_{LOST} IS LOST VOLTS MEASURED IN VOLTS (V) & V_{TPD} IS tpd MEASURED IN VOLTS (V)

EXPERIMENT ① METHOD:

- ~~VOLTAGE~~ TERMINAL POTENTIAL DIFFERENCE WAS MEASURED USING VOLTMETER IN CLOSED CIRCUIT.
- CURRENT THROUGH RESISTOR WAS MEASURED USING AMMETER.
- CURRENT WAS ALTERED BY CHANGING RESISTOR THAT WAS IN SERIES.
- SET UP THIS CIRCUIT:



EXPERIMENT ① RESULTS:

TERMINAL POTENTIAL DIFFERENCE (V)						CURRENT THROUGH RESISTOR (A)					
1	2	3	4	5	MEAN	1	2	3	4	5	MEAN
0.67	0.76	0.74	0.73	0.68	0.72	0.58	0.69	0.68	0.67	0.63	0.65
0.74	0.86	0.85	0.82	0.78	0.81	0.48	0.55	0.55	0.53	0.51	0.52
0.93	1.03	1.02	0.99	0.96	0.99	0.33	0.36	0.36	0.35	0.34	0.35
1.02	1.09	1.09	1.07	1.04	1.06	0.26	0.28	0.28	0.27	0.27	0.27
1.11	1.17	1.18	1.14	1.11	1.14	0.20	0.21	0.21	0.20	0.20	0.20
1.18	1.22	1.23	1.20	1.17	1.20	0.14	0.15	0.15	0.14	0.14	0.14

EXPERIMENT ① UNCERTAINTIES.

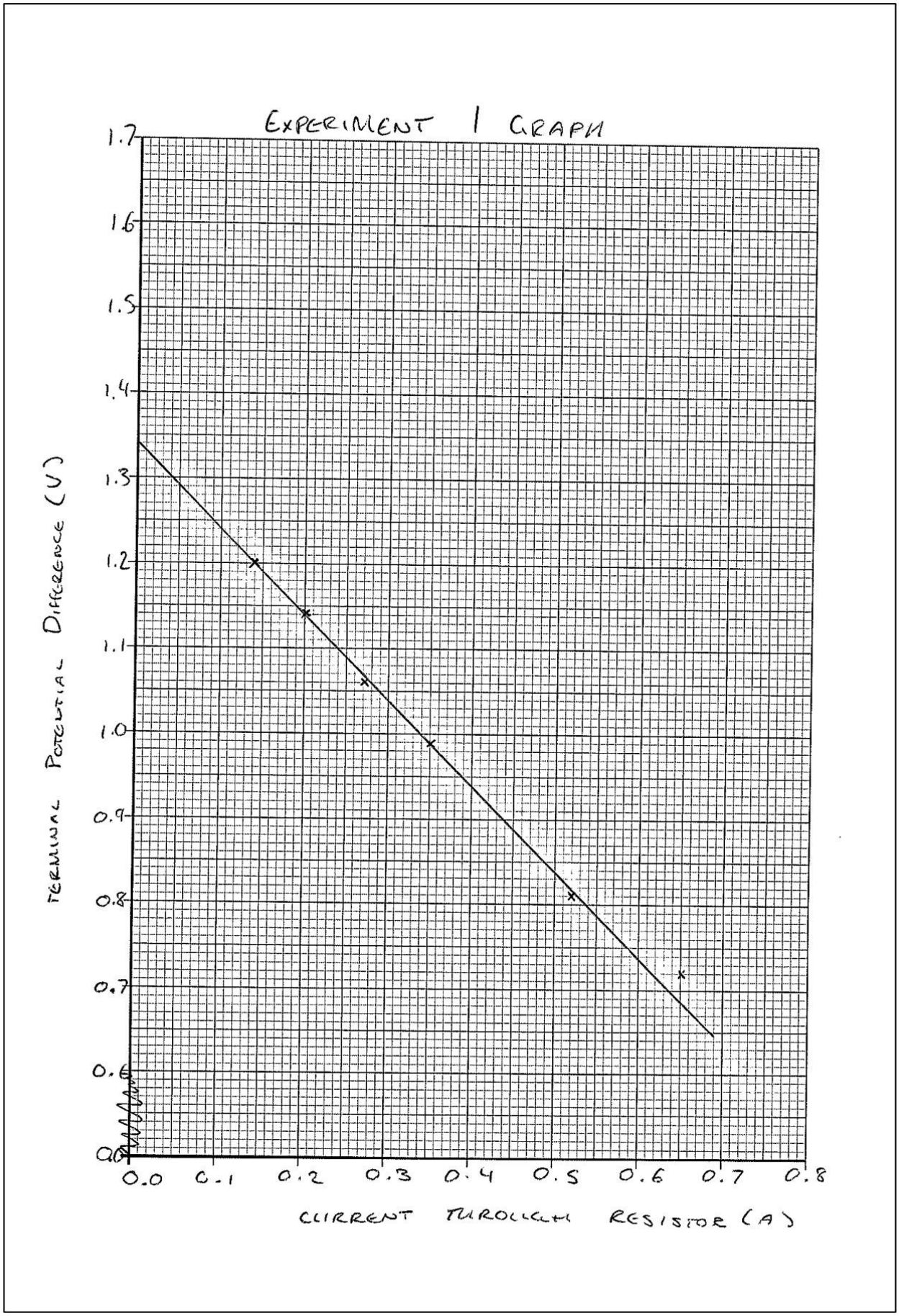
SCALE READING UNCERTAINTIES: VOLTMETER : ±0.01V

AMMETER : ±0.01A

RANDOM UNCERTAINTIES:

MEAN t.p.d. (V)	RANDOM UNCERTAINTIES FOR MEAN t.p.d (V)	MEAN I THROUGH R (A)	RANDOM UNCERTAINTIES FOR MEAN I THROUGH R (A)
0.72	$\frac{0.76-0.68}{5} = 0.016$	0.65	$\frac{0.69-0.58}{5} = 0.022$
0.81	$\frac{0.86-0.74}{5} = 0.024$	0.52	$\frac{0.55-0.48}{5} = 0.014$
0.99	$\frac{1.03-0.93}{5} = 0.020$	0.35	$\frac{0.36-0.33}{5} = 0.030$
1.06	$\frac{1.09-1.02}{5} = 0.014$	0.27	$\frac{0.28-0.26}{5} = 0.004$
1.14	$\frac{1.18-1.11}{5} = 0.014$	0.20	$\frac{0.21-0.20}{5} = 0.002$
1.20	$\frac{1.23-1.17}{5} = 0.012$	0.14	$\frac{0.15-0.14}{5} = 0.002$

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ANALYSIS OF EXPERIMENT (1)

$y = mx + c$ $\quad E = V + IR$

IN TERMS OF EQUATION OF A STRAIGHT LINE.

$V = -r(I) + E$ $\quad \therefore$ y INTERCEPT $= E$ (emf)

AND y GRADIENT $= -r$ (INTERNAL RESISTANCE

$E = 1.34\text{V}$

$$-r = \frac{y_2 - y_1}{x_2 - x_1} \qquad (0.4, 0.94)$$

$$= \frac{1 - 0.94}{0.34 - 0.4} \qquad (0.34, 1.0)$$

$$= -1$$

$$r = 1\,\Omega$$

METHOD OF EXPERIMENT (2)

- RESISTANCE OF RESISTORS WAS MEASURED USING AN OHMMETER
- CURRENT THROUGH R WAS MEASURED USING AMMETER
- RESISTOR IN SERIES WAS CHANGED & CURRENT THROUGH R WAS RECORDED * IN SAME CIRCUIT AS EXPERIMENT 1.

SOURCE 2 RESULTS

RESISTANCE OF R (Ω)	CURRENT THROUGH R (A)						$\frac{1}{I}$ (A^{-1})
	1	2	3	4	5	MEAN	
1.1	0.54	0.55	0.53	0.53	0.60	0.55	1.82
1.6	0.36	0.45	0.44	0.48	0.51	0.45	2.23
3.0	0.32	0.32	0.31	0.34	0.35	0.33	3.05
4.1	0.26	0.26	0.24	0.27	0.27	0.26	3.85
5.8	0.20	0.20	0.18	0.20	0.20	0.20	5.10
8.4	0.14	0.14	0.13	0.15	0.15	0.14	7.04

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EXPERIMENT (2) UNCERTAINTIES

SCALE READING UNCERTAINTIES :

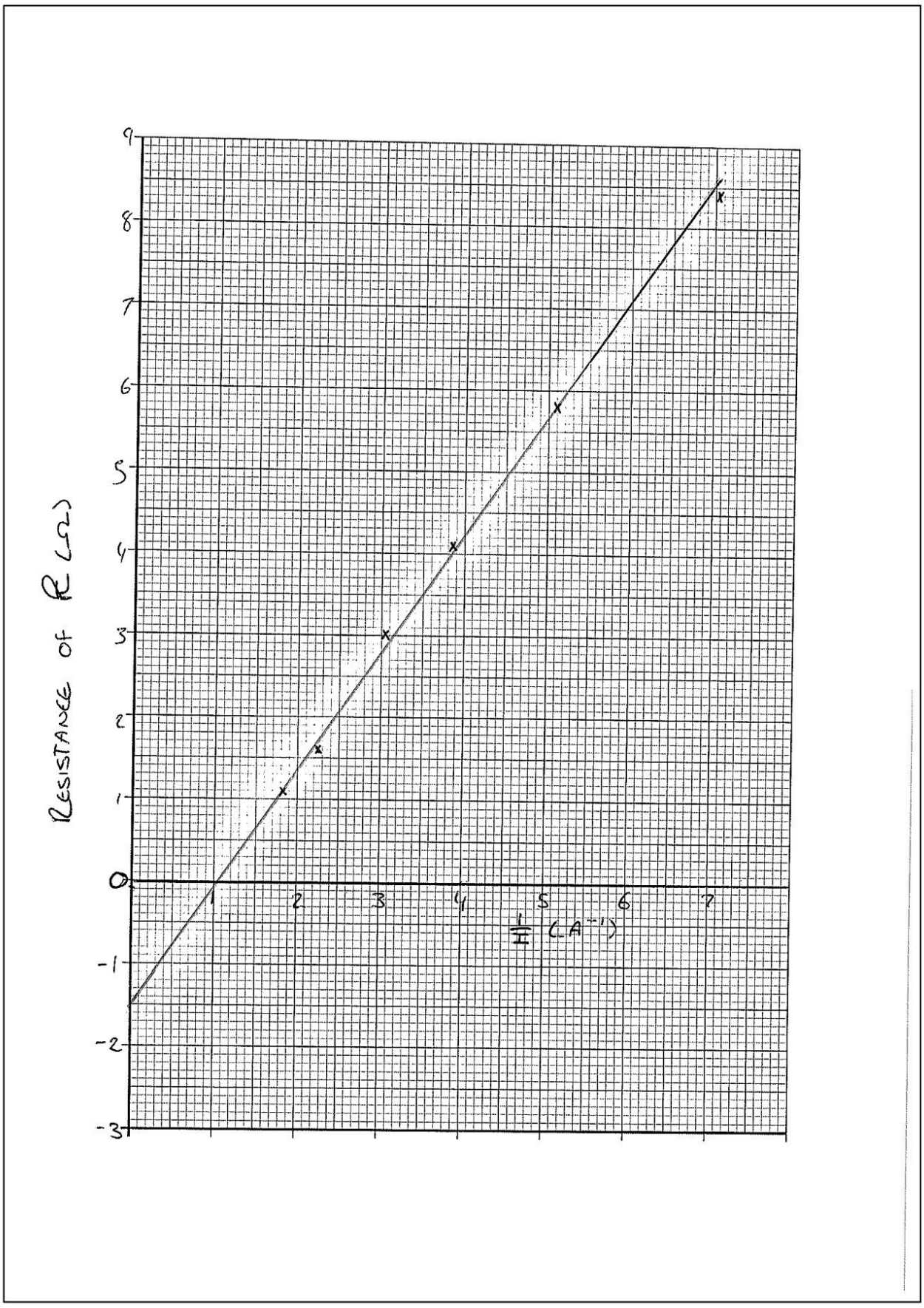
OHMETER : ± 0.10 Ω

AMMETER : ± 0.01 A

RANDOM UNCERTAINTIES :

MEAN CURRENT THROUGH R (A)	RANDOM UNCERTAINTY OF MEAN CURRENT THROUGH R (A)
0.55	$\frac{0.60-0.55}{5} = 0.010$
0.45	$\frac{0.51-0.36}{5} = 0.030$
0.33	$\frac{0.35-0.31}{5} = 0.004$
0.26	$\frac{0.27-0.24}{5} = 0.006$
0.20	$\frac{0.2-0.18}{5} = 0.004$
0.14	$\frac{0.15-0.13}{5} = 0.004$

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ANALYSIS OF EXPERIMENT (2)

$y = mx + c$ IN TERMS OF ~~WITH r~~ $\quad E = I(R + r)$

EQUATION OF A STRAIGHT LINE

$\frac{E}{I} = R + r$

$R = \frac{E}{I} - r$

$R = E \times \frac{1}{I} - r$ $\quad \therefore$ ~~M = E (emf)~~ E(emf) = GRADIENT (m)

AND $-r$ (INTERNAL RESISTANCE = y INTERCEPT)

E(emf) $= \frac{6.5 - 4.5}{5.6 - 4.2}$ $\quad (4.2, 4.5)(5.6, 6.5)$

$= 1.43$ V

$-r = -1.5$

$r = 1.5\ \Omega$

CONCLUSION

THE INTERNAL RESISTANCE OF THE D CELL WAS FOUND TO BE: 1 Ω FROM EXPERIMENT (1)

AND 1.5 Ω FROM EXPERIMENT (2)

EVALUATION

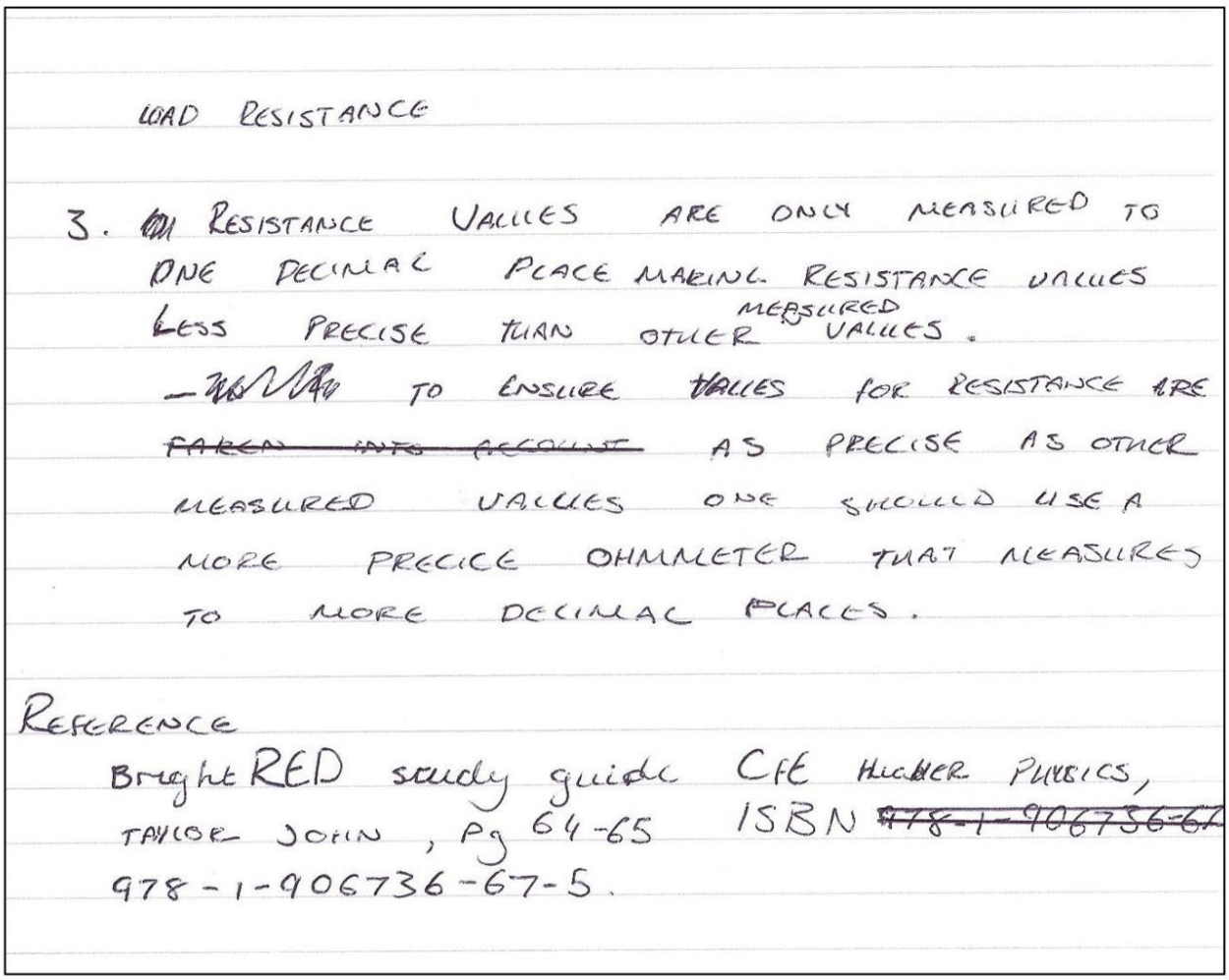
1. THROUGHOUT BOTH EXPERIMENTS ~~RES~~ THE BATTERY HEATED UP, INCREASING THE INTERNAL RESISTANCE.

- IF I WERE TO REPEAT THE EXPERIMENT, I WOULD ALLOW THE BATTERY TIME, INBETWEEN READINGS, TO COOL DOWN.

2. WIRES USED IN EXPERIMENT ALSO HAVE INTERNAL RESISTANCE THAT WAS NOT ACCOUNTED FOR IN THE LOAD RESISTANCE. MEANING THE y INTERCEPT IS HIGHER THAN IT WOULD BE IF r OF WIRES WAS TAKEN INTO ACCOUNT

- TO PREVENT THIS INTERNAL RESISTANCE OF WIRES SHOULD BE MEASURED AND ADDED TO

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LOAD RESISTANCE

3. RESISTANCE VALUES ARE ONLY MEASURED TO ONE DECIMAL PLACE MAKING RESISTANCE VALUES LESS PRECISE THAN OTHER MEASURED VALUES.

~~TO~~ TO ENSURE ~~VALUES~~ FOR RESISTANCE ARE ~~TAKEN INTO ACCOUNT~~ AS PRECISE AS OTHER MEASURED VALUES ONE SHOULD USE A MORE PRECICE OHMMETER THAT MEASURES TO MORE DECIMAL PLACES.

REFERENCE

Bright RED study guide CfE Higher Physics, TAYLOR JOHN, Pg 64-65 ISBN ~~978-1-906736-67-5~~ 978-1-906736-67-5.

End of script 2. **Fill in** grid 2 on your worksheet now.

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5 Script 3: Peak a.c. and d.c. voltage

This assignment investigated how the peak voltage of an a.c. supply compares with its d.c. equivalent. It is 5 pages long.

Read the whole script first. Then **fill in** grid 3 on your worksheet.

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A.C and D.C Voltage

Aim

To find out the relationship between the peak voltage of an a.c supply and its d.c equivalent voltage.

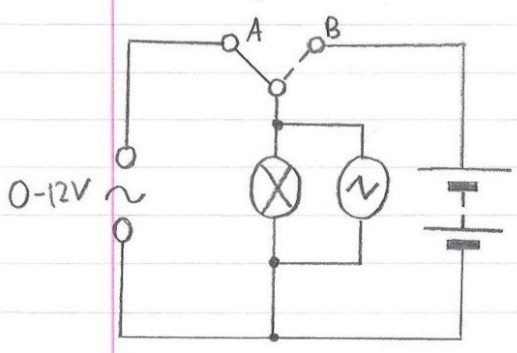
Underlying Physics

Electricity can flow in two different ways:

- Alternating Current (A.C.) - The electrons move forwards then backwards and alternate between them. This results in the voltage always changing. When observed on an Oscilloscope it has a wave shape.

- Direct Current (D.C) - The electrons only flow forward so have a constant voltage. When DC is observed on an Oscilloscope a straight line is seen.

Method



The D.C. Voltage was measured with a voltmeter. The peak A.C voltage was measured with an oscilloscope in position A.

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A.C and D.C Voltage

Results

Number of Cells	d.c. voltage (V)					random uncertainty (V)
	1	2	3	4	mean	
1	1·35	1·34	1·32	1·32	1·33	0·008
2	2·66	2·64	2·61	2·60	2·63	0·02
3	3·70	4·37	4·37	4·38	4·21	0·17
4	4·89	5·70	5·70	5·69	5·50	0·20
5	7·05	7·03	7·05	6·93	7·02	0·03

Scale Reading Uncertainty ±0·01V

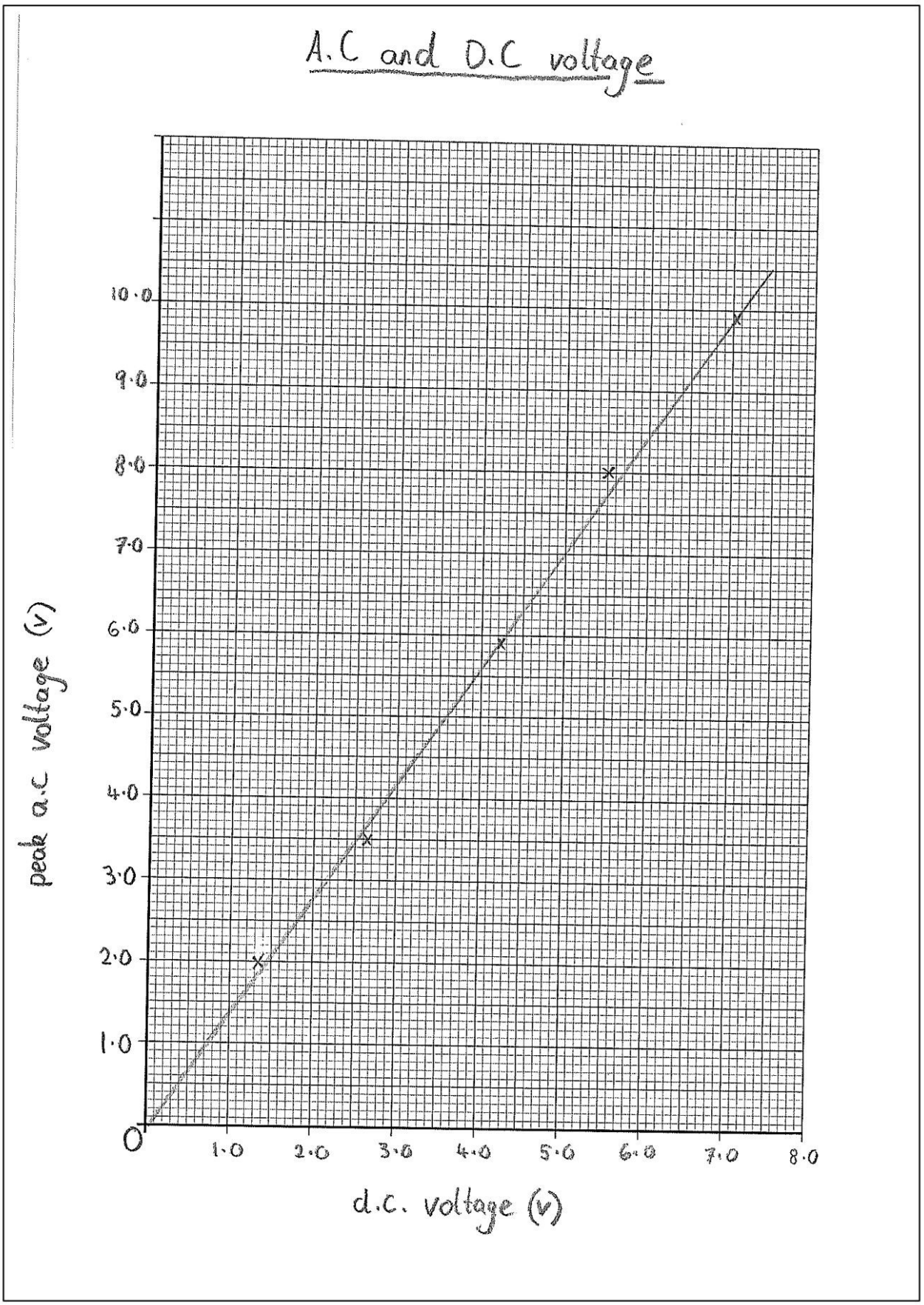
Number of Cells	peak a.c voltage (V)					random uncertainty (V)
	1	2	3	4	mean	
1	2·0	2·0	2·0	2·0	2·0	0
2	4·0	3·6	3·2	3·3	3·5	0·2
3	4·8	5·9	6·8	6·2	5·9	0·5
4	7·6	8·1	8·1	8·0	8·0	0·1
5	10	10	9·9	9·8	9·9	0·1

$\text{mean} = \frac{4·0 + 3·6 + 3·2 + 3·3}{4}$ $\text{random uncertainty} = \frac{4·0 - 3·2}{4}$

$= 3·5V$ $= 0·2V$

Scale Reading Uncertainty ±0·5V

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A.C and D.C voltage

Analysis

$(5.0, 7.0)$ $(2.5, 3.5)$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{7.0 - 3.5}{5.0 - 2.5}$$

$$= 1.4$$

The gradient of the graph plotted is 1.4 which shows that the peak a.c voltage is a factor of 1.4 higher than its d.c equivalent.

Second Source

Reference 1

	V_m (volts)	V_{dc} (volts)
lamp 1	12.0	8.5
lamp 2	3.0	2.1
lamp 3	2.0	1.4

$$\frac{V_m}{V_{dc}} = \frac{12.0}{8.5} = 1.4$$

$$\frac{V_m}{V_{dc}} = \frac{3.0}{2.1} = 1.4$$

$$\frac{V_m}{V_{dc}} = \frac{2.0}{1.4} = 1.4$$

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A.C and D.C voltage

Conclusion

The peak voltage of an a.c supply is a factor of 1.4 higher than its d.c equivalent voltage.

Evaluation

An LDR could have been used to measure the irradiance of the bulb which would improve the accuracy of the experiment.

The second source didn't have any uncertainties listed so the precision of the experiment is unknown.

The uncertainties of the data collected was higher than 3% for some results so the data was not as precise as it could have been.

Reference 1 - Book

Higher Core Physics Serond Edition, Geoff. Cackett, Jim Lowrie, Alastair Steven, Page 91,
ISBN 0 19 914324 2

End of script 3. **Fill in** grid 3 on your worksheet now.

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6 Script 4: Irradiance and distance

This assignment investigated how the irradiance from a point source changes with distance. It is 10 pages long.

Read the whole script first. Then **fill in** grid 4 on your worksheet.

Candidate 4 evidence, Physics Higher Assignment 2023-24, Understanding Standards. © Qualifications Scotland (SQA).

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The inverse Square Law of Irradiance

Aim: To determine the relationship between the distance from a point source and the irradiance at that point.

Underlying Physics:

The inverse square law is a principle which states that light propagates evenly in space.

A point source is used when investigating the inverse square law of irradiance as it has no limit to its range. The smaller a source of light is, the more it behaves like a point source. An example of a point source would be a star. A point source emits light equally in all directions.
The inverse square law of irradiance states that as the distance from a point source increases, its irradiance decreases. This is because there is a lesser number of photons hitting the same point each second.

There are several equations linking irradiance and distance:

$I = \frac{k}{d^2}$ $I \propto \frac{1}{d^2}$ $I_1 d_1^2 = I_2 d_2^2$

$I = \frac{P}{A}$

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Irradiance is known as the power per metre squared. It is measured in (Wm^{-2}). The area can be calculated using $4\pi r^2$ (m^2). This proves $I = \frac{P}{A}$ *

Irradiance is inversely proportional to distance square. As Irradiance multiplied by distance squared equals a constant, we can prove that $I_1 d_1^2 = I_2 d_2^2$, aswell as $I = \frac{k}{d^2}$.

Where : I - irradiance (Wm^{-2})
d - distance (m^2)
k - Constant

* Where I = irradiance (Wm^{-2})
P = Power (W)
A = Area (m^2)

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Experimental Method:

A bulb was connected to a power supply and placed opposite from a light sensor.* The light sensor was connected to the ALba Saftware on the computer. The ~~voltage of the bulb was recorded by~~ bulb was then switched on. The ALba Software recorded the irradiance from the bulb. A background reading of irradiance was recorded first to correct the results.

* The distance was measured using a ruler.

Distance between bulb and light sensor (cm)	$\frac{1}{d^2}$ (cm^{-2})	Irradiance ($W cm^{-2}$)				
		1	2	3	4	mean
20	2.5×10^{-3}	0.120	0.124	0.114	0.119	0.119
22	2.07×10^{-3}	0.101	0.103	0.101	0.102	0.102
24	1.74×10^{-3}	0.091	0.090	0.086	0.085	0.088
26	1.48×10^{-3}	0.079	0.076	0.075	0.077	0.077
28	1.28×10^{-3}	0.068	0.067	0.065	0.068	0.067

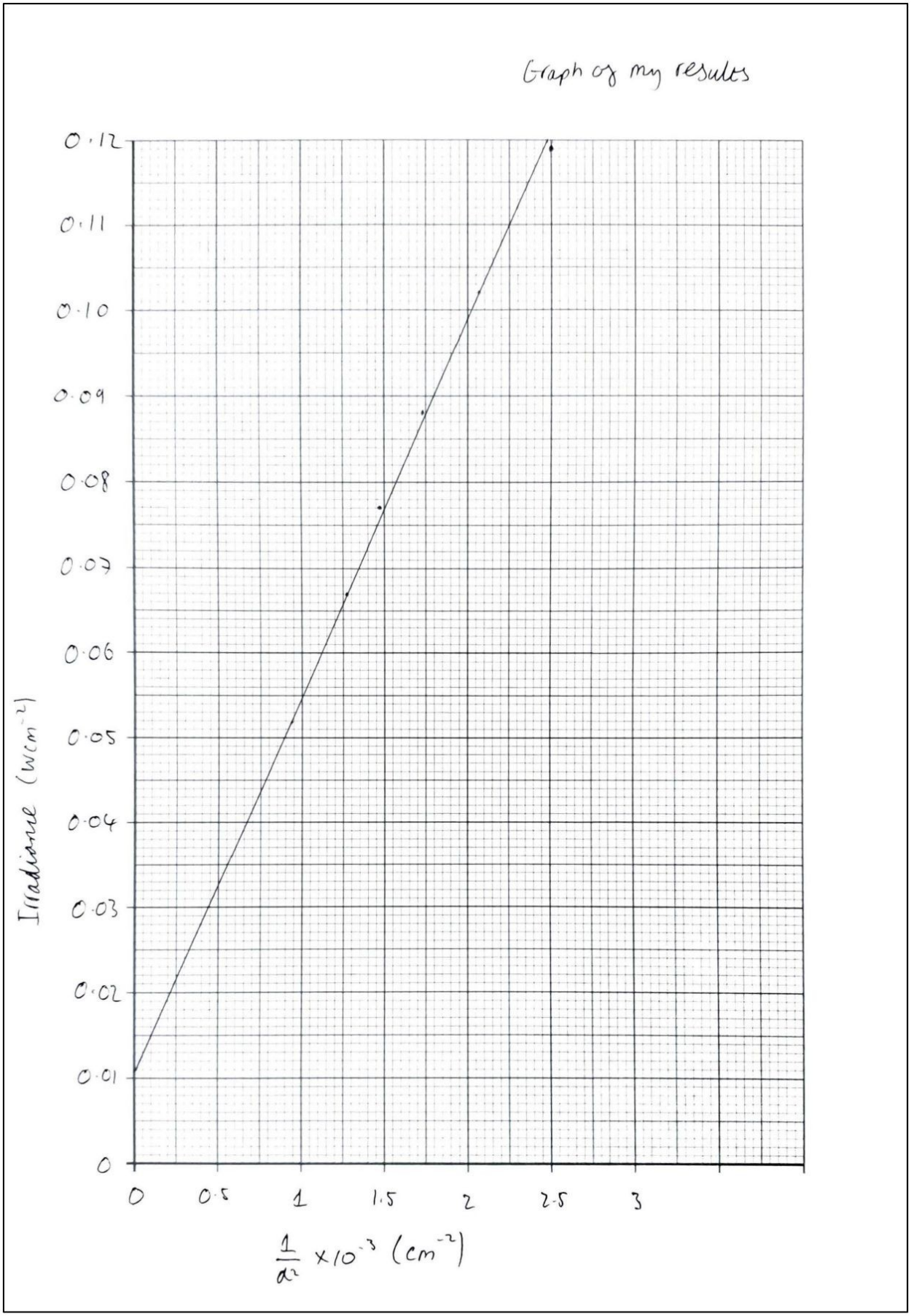
$$\frac{1}{d^2} = \frac{1}{20^2}$$

$$= \underline{\underline{2.5 \times 10^{-3}}}$$

$$\text{mean} = \frac{0.068 + 0.067 + 0.065 + 0.068}{4}$$

$$= 0.067$$

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Uncertainties:

Scale Reading uncertainties

The Scale reading uncertainty for the ruler was ± 0.05cm.

The Scale reading uncertainty for the alba Software was $\pm 0.001\ Wcm^{-2}$

Random uncertainty,

For 20 cm $\quad \frac{0.124 - 0.114}{4}$

$= \pm 2.5 \times 10^{-3}\ Wcm^{-2}$

For 22 cm $\quad \frac{0.103 - 0.101}{4}$

$= \pm 5 \times 10^{-4}\ Wcm^{-2}$

For 24 cm $\quad \frac{0.091 - 0.085}{4}$

$= \pm 1.5 \times 10^{-3}\ Wcm^{-2}$

For 26 cm $\quad \frac{0.079 - 0.075}{4}$

$= \pm 1 \times 10^{-3}\ Wcm^{-2}$

For 28 cm $\quad \frac{0.068 - 0.065}{4}$

$= \pm 7.5 \times 10^{-4}\ Wcm^{-2}$

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Analysis:

Gradient of graph $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{0.09 - 0.015}{(1.8 \times 10^{-3}) - (0.1 \times 10^{-3})}$

$= 44.1176\ldots$

$= 44.1$

Constant = gradient of graph

Constant = 44.1

$I = \frac{k}{d^2}$ $I = \frac{44.1}{d^2}$

$I d_2 = k$

20cm- $0.119 \times 20^2 = 47.6$

22cm- $0.102 \times 22^2 = 49.4$

24cm- $0.088 \times 24^2 = 50.7$

26cm- $0.077 \times 26^2 = 52.1$

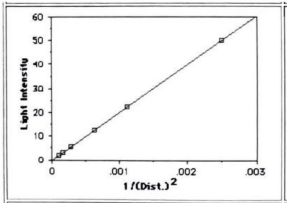
28cm $0.067 \times 28^2 = 52.5$

As there were uncertainties in my results, my graph proves that $I = \frac{k}{d^2}$. If the uncertainties were reduced I would have had more accurate values.

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The data below was gathered while observing the manner in which light intensity dropped off as the student moved away from a light source. Use one or more of the four methods of analyzing the data to see if you can discover the mathematical relationship. If you think you have the form of the proportion, then go ahead by clicking the method you used to do the analysis.

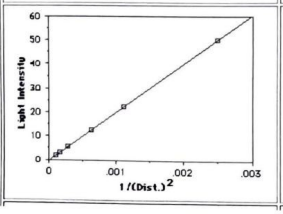
Distance cm	Light Intensity units
20	50
30	22.2
40	12.5
60	5.5
80	3.1
100	2.0



By plotting the reciprocals of the distances squared versus the intensities, we now have a straight line graph. This is a solid indication of the type of proportion that is at work in this lab.

From the graph we can immediately write:

$$I = k\,(1/d^2)$$



The mathematical proportion derived from the straight line states that the intensity (I) is directly proportional to the square of the reciprocals of the distance ($1/d^2$).

A proportion of this type is called an "**Inverse Square**" proportion.

Conclusion:

To conclude, through my experimental results, I calculated that $I \propto \frac{1}{d^2}$

Evaluation:

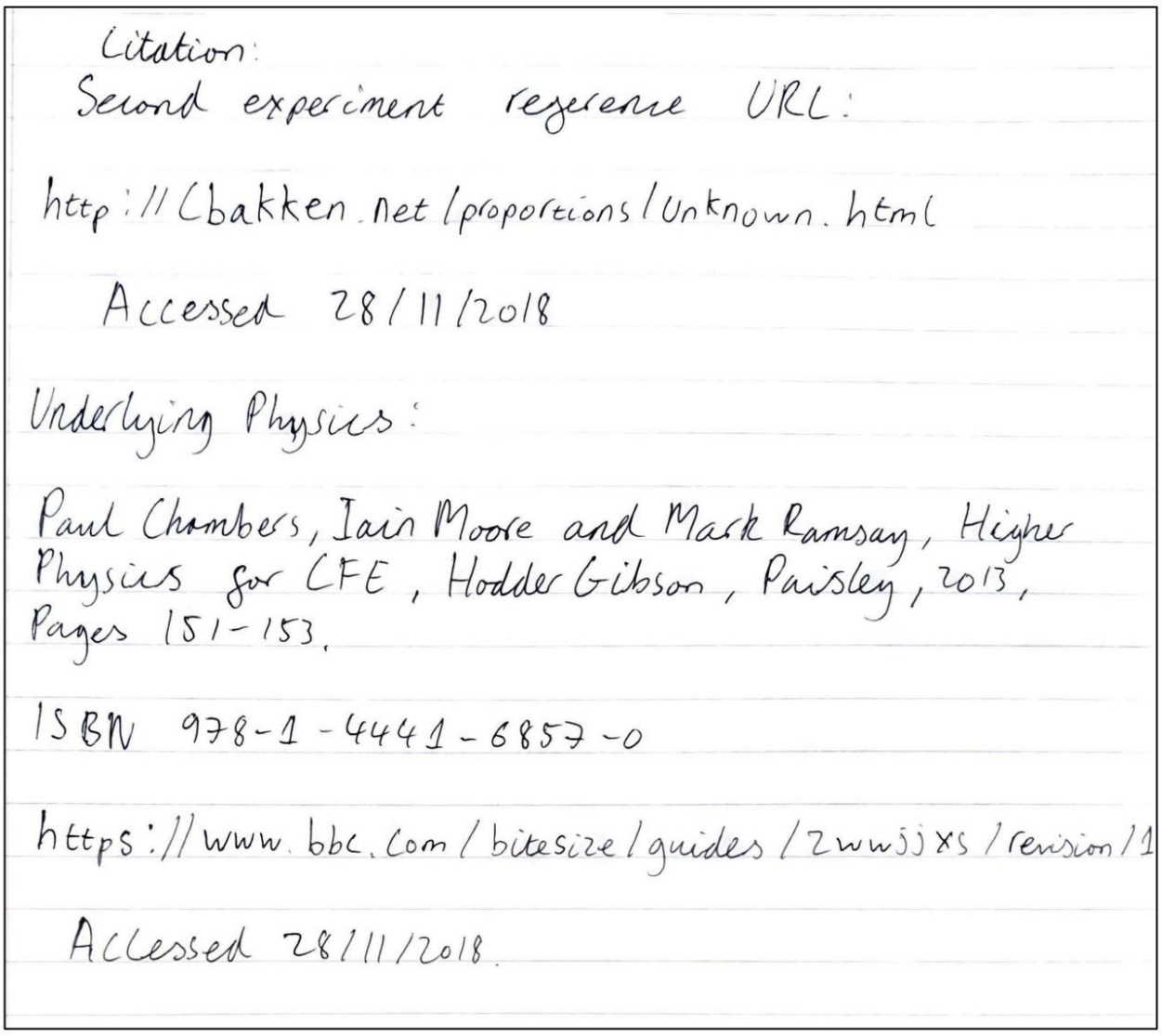
My experimental results were reliable as they followed the same trend as my second data source.

To decrease my random uncertainty I could have had a greater number of repeats for my experiment. This would improve the accuracy of my results.

The best fit line on my graph should have cut through the origin. This didn't happen as the light level in the room changed. To improve my results I could have kept the background light level constant, or taken the background light level before every result, this would allow me to correct my results. Allowing the graph to cut through the origin.

To reduce my scale reading uncertainty I could have used a ruler with a smaller unit of values. This would improve the accuracy of my results.

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Citation:
Second experiment reference URL:

http://bakken.net/proportions/Unknown.html

Accessed 28/11/2018

Underlying Physics:

Paul Chambers, Iain Moore and Mark Ramsay, Higher Physics for CFE, Hodder Gibson, Paisley, 2013, Pages 151-153.

ISBN 978-1-4441-6857-0

https://www.bbc.com/bitesize/guides/zwwjjxs/revision/1

Accessed 28/11/2018.

End of script 4. **Fill in** grid 4 on your worksheet now.

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7 Script 5: Refractive index of water

This assignment investigated the refractive index of water. It is 6 pages long.

Read the whole script first. Then **fill in** grid 5 on your worksheet.

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Verifying the Refractive Index of Water

Aim: To verify the refractive index of water

Underlying Physics: The refractive index of a material is the measure of how light is bent when entering a new medium. When the light is coming directly into the new material from a vacuum it is known to be the absolute refractive index of that material. As the refractive index increases so too does the degree of the angle of refraction.

When plotting a graph of the angle of incidence against the angle of refraction it will appear that the relationship between the two is not clear and doesn't produce a straight line through the origin. Therefore to plot a graph correctly which will show a relationship of the two being directly proportional, the sine of the angle of incidence must be plotted against the sine of the angle of refraction, which should produce a straight line through the origin. The straight line through the origin shows that the two axes are directly proportional:

$$\sin\theta_1 \propto \sin\theta_2.$$

The gradient of the straight line should provide an accurate value of the refractive index of the material. When light travels into the new medium the ratio of $\sin\theta_1$ to $\sin\theta_2$ is a constant unless the light enters the medium at 90°.

The absolute refractive index, n, of a material is the ratio

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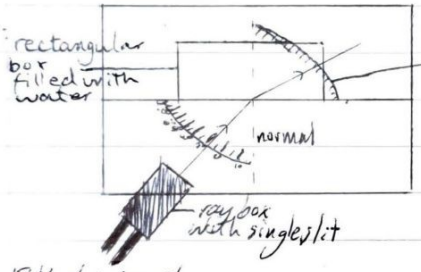
$\frac{\sin\theta_1}{\sin\theta_2}$ where θ_1 is in a vacuum and θ_2 is in the medium. This gives Snell's Law:

$$n = \frac{\sin\theta_1}{\sin\theta_2}$$

where: n = absolute refractive index of the medium
$\sin\theta_1$ = angle of incidence in a vacuum
$\sin\theta_2$ = angle of refraction in the medium.

In practice, we can use $n = \frac{\sin\theta_{air}}{\sin\theta_{medium}}$ (1)

Method



To 12V d.c. supply.

During my experiment I moved the ray box from 10° through to 70°. Each time I changed the angle of incidence I recorded the angle of refraction shown on the protractor. I repeated these steps until I had all my data.

Results

Angle of Incidence θ_1 (°)	Angle of Refraction, θ_2 (°)					Scale Reading Uncertainty	Random Uncertainty	Dominant Uncertainty	Percentage Uncertainty
	1	2	3	4	Mean				
10	7	9	8	8	8	0.5	0.5	0.5	6.25%
20	16	14	15	15	15	0.5	0.5	0.5	3.33%
30	23	23	24	22	23	0.5	0.5	0.5	2.17%
40	27	28	29	28	28	0.5	0.5	0.5	1.79%
50	36	33	35	34	34.5	0.5	0.75	0.75	2.17%
60	38	39	40	39	39	0.5	0.5	0.5	1.28%
70	42	45	44	45	44	0.5	0.75	0.75	1.70%

percentage uncertainty

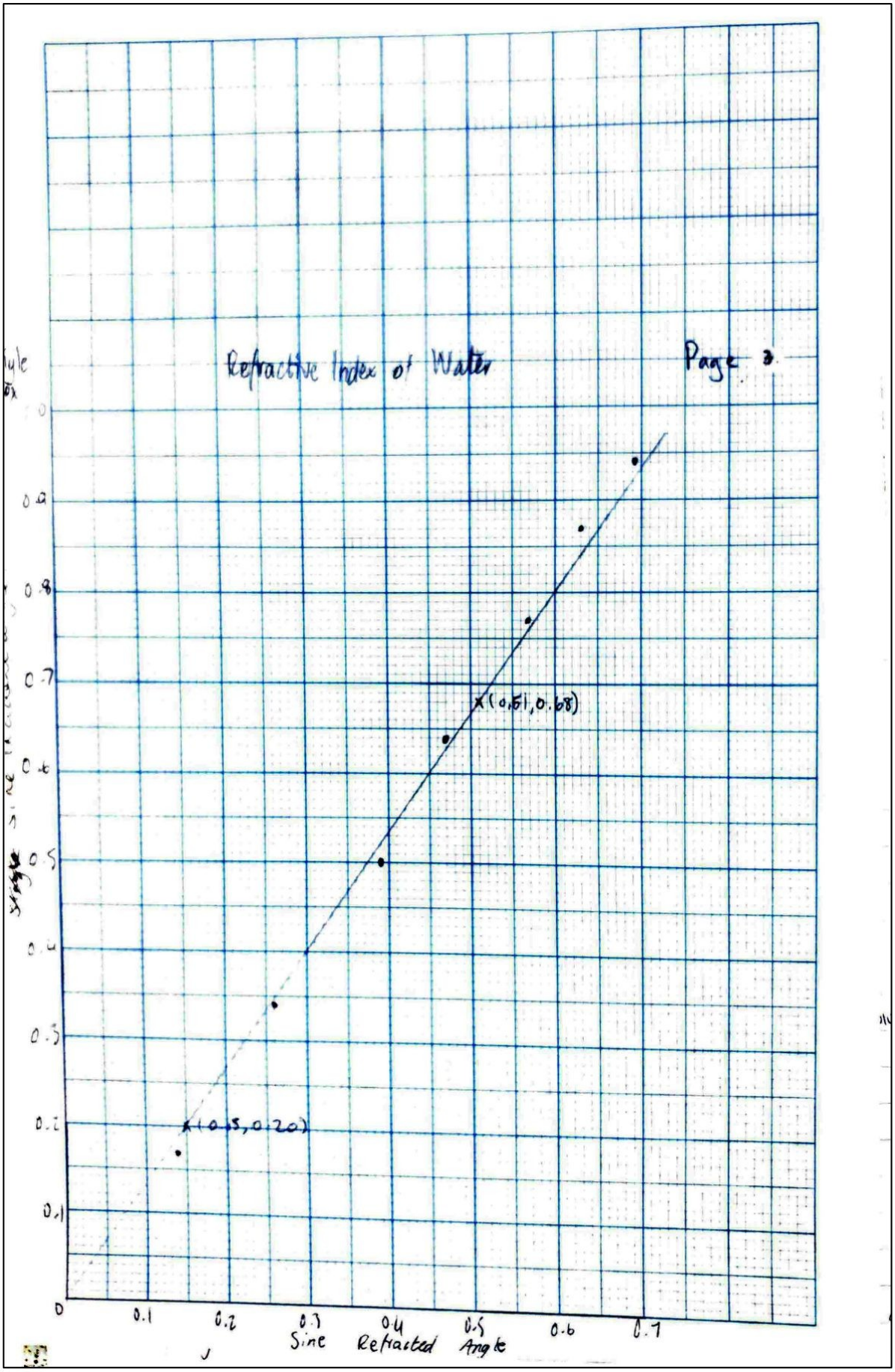
$7+9+8+8 = \frac{32}{4} = 8$

$9-7 = \frac{2}{4} = 0.5$

$\%\ \text{of } 8 = \frac{0.5}{8} \times 100$

$= 6.25\%$

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Graph data

$\sin\theta_1$	$\sin\theta_2$
0.17	0.14
0.34	0.26
0.50	0.39
0.64	0.47
0.77	0.57
0.87	0.63
0.94	0.69

$\sin 10 = 0.1736...$

Uncertainties

Scale Reading = ±0.5° for protractor.

Random Uncertainty –

Angle of Refraction, θ_2 (°)				Random Uncertainty
1	2	3	4	
7	9	8	8	0.5
16	14	15	15	0.5
23	23	24	22	0.5
27	28	29	28	0.5
36	33	35	34	0.75
38	39	40	39	0.5
42	45	44	45	0.75

$4 \cdot 7 = \frac{2}{4} = 0.5$

Analysis

$(x^1, y^1) = (0.15, 0.20)$ $(x^2, y^2) = (0.51, 0.68)$

$m = \frac{y^2 - y^1}{x^2 - x^1}$

$= \frac{0.68 - 0.20}{0.51 - 0.15}$

$= \frac{0.48}{0.36}$

$= 1.333...$

$= 1.33$

Second Data Source (page 5)

$(x^1, y^1) = (0.1397459, 0.1827101)$ $(x^2, y^2) = (0.615120, 0.890109)$

$m = \frac{y^2 - y^1}{x^2 - x^1}$

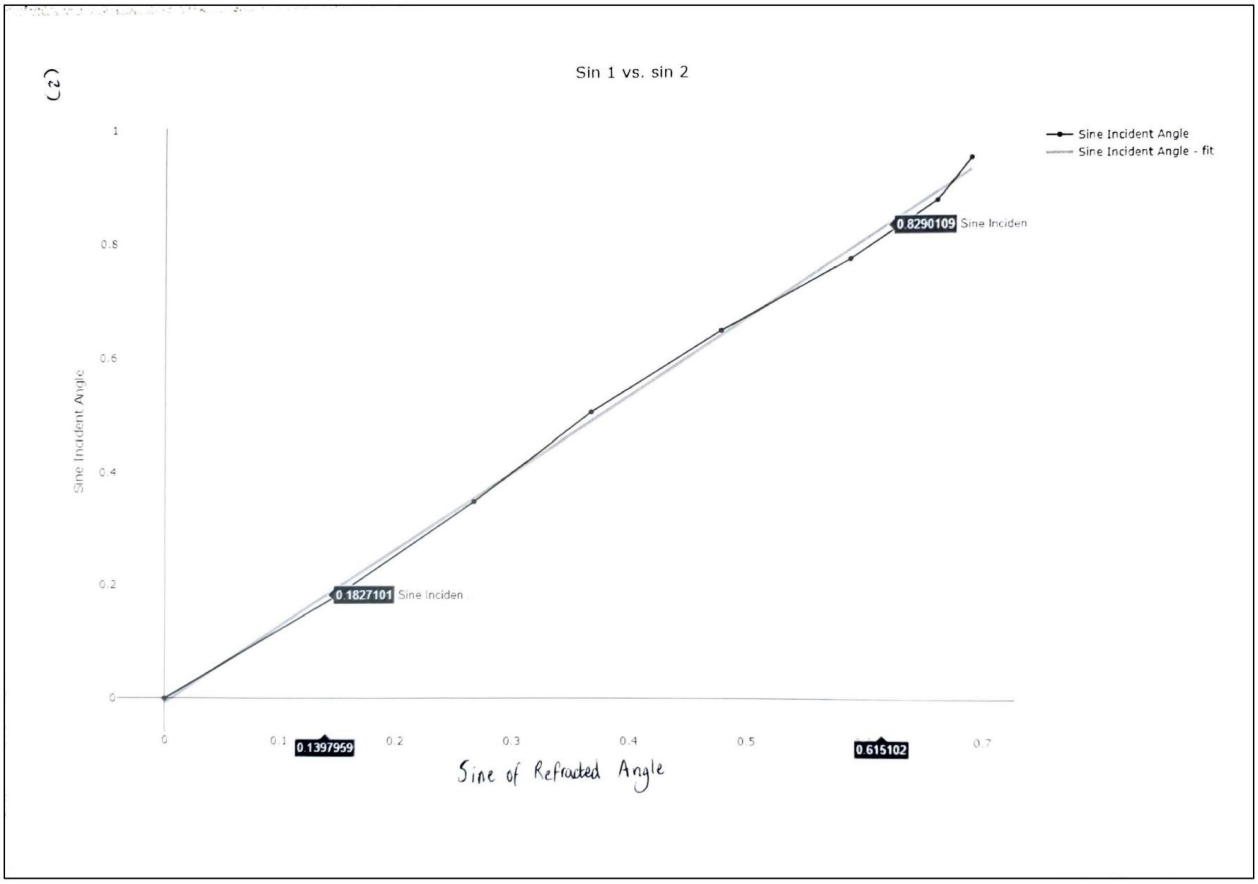
$= \frac{0.890109 - 0.1827101}{0.615120 - 0.1397459}$

$= 1.3507...$

$= 1.36$

My experimental data provided a value closer to the correct value of

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(c)
Sin 1 vs. sin 2
Sine Incident Angle
Sine Incident Angle - fit
Sine Incident Angle
1
0.8
0.6
0.4
0.2
0
0.1
0.2
0.3
0.4
0.5
0.7
0.1827101
0.8290109
0.1397959
0.615102
Sine of Refracted Angle

the refractive index of water

Conclusion

After plotting a graph of $\sin\theta_1$ against $\sin\theta_2$, I calculated the gradient of the graph which should give me the value of refractive index my data provides. The value calculated from my graph was 1.3326251, and the correct value for the refractive index of water should be 1.33. This shows that when allowing uncertainties I have therefore verified the refractive index of water.

Evaluation

When I was carrying out my experiment I didn't check to see if the water was room temperature, to improve my experiment I would ensure that the water is at room temperature when taking all my data.

The box that I used to contain the water was a rectangle and when the refracted ray of light neared the corners of the box the protractor reading became difficult to read. To improve the accuracy of my readings I could use a semi-circular box to contain the water as there will be no square edges to potentially make it more difficult to read the protractor value.

To further improve my experimental data I could repeat each angle of incidence a few more times to reduce the random uncertainty in my results.

References

1 - Bright Red Study Guides CFE Higher Physics, John Taylor, page 108, ISBN: 978-1-906736-67-5

2 - https://plot.ly/~zameliaII/5.embed November 2018

End of script 5. **Fill in** grid 5 on your worksheet now.

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8 Answers

These are the marker's marks, from the Understanding Standards commentary. **Mark the scripts yourself before you look.**

Item	Section	Max	Script 1	Script 2	Script 3	Script 4	Script 5
1	Aim	1	1	1	1	1	1
2	Underlying physics	3	1	3	0	2	1
3a	Method summary	1	1	0	0	0	1
3b	Raw data	1	1	1	1	1	1
3c	Data table	1	0	1	1	1	1
3d	Second data	1	1	1	0	1	1
3e	Citation and reference	1	0	0	1	0	1
4a	Scales	1	1	1	1	1	1
4b	Labels and units	1	0	1	1	1	1
4c	Plotting and line	1	1	1	1	1	0
5	Uncertainties	2	1	1	1	2	1
6	Analysis	1	1	1	1	0	0
7	Conclusion	1	0	1	1	0	1
8	Evaluation	3	2	3	1	1	2
9	Title and structure	1	0	1	1	1	1
	Total	**20**	**11**	**17**	**12**	**13**	**14**

Focus task answers

Task 1. Script 4: statement 1 is not enough for a mark. Statement 2 says more repeats would improve accuracy, but fewer random uncertainties improve the precision, not the accuracy (0). Statement 3, measuring the background light and subtracting it, reduces a systematic uncertainty (1 mark: sources of uncertainty). Statement 4 asks for a 1 mm ruler, which was already used (0). Script 5: the water temperature statement does not say that temperature affects the refractive index (0). The rectangular box made the protractor hard to read, and a semicircular box would fix it (1 mark: limitations of equipment). More repeats would reduce the random uncertainty, due to the variation in the results (1 mark: adequacy of repeated readings).

Task 2. Script 3 scored 0 out of 3: it shows no understanding of any relevant physics at Higher level. It describes a.c. and d.c. in a way a National 5 pupil could. Higher physics it could have used: peak and r.m.s. voltage, V(peak) = √2 × V(r.m.s.), and what the r.m.s. value means. Its gradient of 1.4 is close to √2. Script 2 scored 3 out of 3, using E = V + Ir and explaining terminal p.d. and lost volts.
Task 3. Script 1's aim is to determine Planck's constant, h. Its conclusion gives no value, so it does not answer the aim (7: 0). Script 5 states its value for the refractive index of water (7: 1). A better conclusion for script 1 states the value of h found from the gradient, with its unit, J s.
Task 4. Script 4's line does not pass through the origin. So the further analysis is invalid (6: 0), and so is a conclusion of direct proportion (7: 0). A straight line that misses the origin shows a linear relationship, not direct proportion.
Task 5. Script 1 puts the full URL next to its second source. A URL cannot be a citation (3e: 0). Script 2 has a (1) in the underlying physics, but the (1) is not repeated with the reference (3e: 0). Script 3 cites its book beside the data and lists it later (3e: 1). Script 4 lists its sources but has no citation (3e: 0).
Task 6. Script 1 says what was measured and how, and its circuit diagram shows the voltmeter (3a: 1). Script 3 gives no indication of what is being done in the experiment (3a: 0). Script 4 says what was measured and how, but not what was altered (3a: 0). Your two sentences need what was changed (the distance), what was measured (the irradiance) and the instruments (a light sensor and a ruler).

The full commentary follows. Read the marker's reasons for each section.

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Commentary on candidate evidence

The candidate evidence has achieved the following marks for each section of the assignment.

Candidate 1

1 Aim

The candidate was awarded **1 out of 1 mark** because their aim describes clearly the purpose of the investigation.

2 Underlying physics

The candidate was awarded **1 out of 3 marks** because they demonstrate a limited understanding of the underlying physics relevant to the aim. There is additional underlying physics at the end of the report, but this does not alter the overall judgement of the candidate's understanding. The candidate could have described the operation of LEDs and could have given the equation relating switch on voltage and Planck's constant.

3 Data collection and handling

The candidate was awarded **3 out of 5 marks**. The marks were awarded as follows:

3(a) **1 out of 1 mark** was awarded because the method contains details of what is to be measured and how the experiment was carried out. The circuit diagram adds clarity, for example it shows the voltage was measured using a voltmeter.

3(b) **1 out of 1 mark** was awarded because the candidate has sufficient data. The range of colours of LED is appropriate to the aim and the candidate has repeated measurements of the dependent variable for each value.

3(c) **0 out of 1 mark** was awarded because there is no unit for the QV column so the mark cannot be awarded.

3(d) **1 out of 1 mark** was awarded because the table of data from the internet source is relevant to the candidate's experiment.

3(e) **0 out of 1 mark** was awarded because the candidate has given the full URL for the reference next to the second source. This cannot be used as a citation.

4 Graphical presentation

The candidate was awarded **2 out of 3 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because the candidate has drawn a scatter graph with suitable scales.

4(b) **0 out of 1 mark** was awarded because there is no unit for QV, so the mark is not awarded.

4(c) **1 out of 1 mark** was awarded because the candidate has plotted the points accurately and drawn an acceptable line of best fit.

5 Uncertainties

The candidate was awarded **1 out of 2 marks** because the reading uncertainties are plausible. The units are missing from the random uncertainties.

6 Analysis

The candidate was awarded **1 out of 1 mark** because they have correctly calculated the gradient of their line and stated its significance.

7 Conclusion

The candidate was awarded **0 out of 1 mark** because the aim is to determine h, but no value is given in the conclusion.

8 Evaluation

The candidate was awarded **2 out of 3 marks** because they make three evaluative statements, two of which can be awarded a mark. The first statement compares their data to the internet data but is insufficient to be awarded a mark because they have not stated that this could have reduced the random uncertainty. In the second statement the candidate correctly identifies a systematic uncertainty in their experimental data. In the third statement the candidate suggests an improvement to their procedure.

9 Structure

The candidate was awarded **0 out of 1 mark** because 'Higher Physics Assignment' is not an informative title.

Overall

The candidate was awarded a total of **11 out of 20 marks**.

Candidate 2

1 Aim

The candidate was awarded **1 out of 1 mark** because their aim describes the purpose of the investigation.

2 Underlying physics

The candidate was awarded **3 out of 3 marks** because they demonstrate a good understanding of the underlying physics. The underlying physics written by the candidate at the beginning does not relate to the practical work that the candidate has carried out however the sections labelled 'analysis' both contain more underlying physics relevant to their aim.

3 Data collection and handling

The candidate was awarded **3 out of 5 marks**. The marks were awarded as follows:

3(a) **0 out of 1 mark** was awarded because although two experimental procedures have been described, neither meets the standard required. For method 1, the circuit diagram is incorrect/incomplete. If the circuit diagram had been complete for method 1, then a mark could have been given. For method 2, the summary is insufficient to be awarded the mark.

3(b) **1 out of 1 mark** was awarded because for method 1 there is sufficient data.

3(c) **1 out of 1 mark** was awarded because for both experiments the tables have appropriate headings and units. All the mean and derived values are correctly calculated.

3(d) **1 out of 1 mark** was awarded because data from the second experiment is sufficient and relevant to the aim of the investigation.

3(e) **0 out of 1 mark** was awarded because although a source is referenced, the citation is incorrect. There is a (1) in the underlying physics, but this is not repeated with the reference.

4 Graphical presentation

The candidate was awarded **3 out of 3 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because graph 1 has correct axis and scales. Graph 2 has correct axis and scales.

4(b) **1 out of 1 mark** was awarded because graph 1 has correct units and labels. Graph 2 has correct units and labels.

4(c) **1 out of 1 mark** was awarded because graph 1 has correctly plotted points and an acceptable line of best fit. Graph 2 has correctly plotted points and an acceptable line of best fit. Either graph would be awarded 3 marks.

5 Uncertainties

The candidate was awarded **1 out of 2 marks** because experiment 1 has plausible scale reading uncertainties but there is an error in calculating the random uncertainty in the third value for current. Experiment 2 has plausible scale reading uncertainties (ignoring the additional 0 in the ohmmeter uncertainty) but there is an error in calculating the third random uncertainty. Either of the treatment of uncertainties would be awarded 1 mark.

6 Analysis

The candidate was awarded **1 out of 1 mark** because the analysis of experiment 1 is correct. The gradient is correctly calculated, and its significance stated. The analysis of the second graph, to find the internal resistance, is also correct. Either analysis would be awarded the mark.

7 Conclusion

The candidate was awarded **1 out of 1 mark** because the conclusion answers the aim. The candidate has arrived at two different values for the internal resistance and both values are stated.

8 Evaluation

The candidate was awarded **3 out of 3 marks** because they have made 3 correct evaluative statements. The first statement identifies an issue with conducting the experiment and offers a valid remedy. The second statement identifies an issue and suggests a solution to the issue. The third statement identifies a possible improvement to the experimental procedure.

9 Structure

The candidate was awarded **1 out of 1 mark** because the report is clear and concise and has an informative title.

Overall

The candidate was awarded a total of **17 out of 20 marks**.

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Candidate 3

1 Aim

The candidate was awarded **1 out of 1 mark** because their aim describes the purpose of the investigation.

2 Underlying physics

The candidate was awarded **0 out of 3 marks** because they have not demonstrated an understanding of any relevant physics at Higher level.

3 Data collection and handling

The candidate was awarded **3 out of 5 marks**. The marks were awarded as follows:

3(a) **0 out of 1 mark** was awarded because the summary lacks enough detail to be awarded the mark. The candidate has given no indication of what is being done in the experiment.

3(b) **1 out of 1 mark** was awarded because the candidate has sufficient data. The range of readings is appropriate to the aim and the candidate has repeated measurements of the dependent variable for each value.

3(c) **1 out of 1 mark** was awarded because the candidate's tables have correct units and headings. The mean values are correctly calculated and tabulated.

3(d) **0 out of 1 mark** was awarded because the candidate has used a literature source that has only three data sets. This is not sufficient to be awarded the mark.

3(e) **1 out of 1 mark** was awarded because the source is cited beside the data and referenced later in the report.

4 Graphical presentation

The candidate was awarded **3 out of 3 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because the candidate has drawn a scatter graph with suitable scales.

4(b) **1 out of 1 mark** was awarded because the axes of the graph have suitable labels and units.

4(c) **1 out of 1 mark** was awarded because the candidate has plotted the points accurately and drawn an acceptable line of best fit.

5 Uncertainties

The candidate was awarded **1 out of 2 marks** because the scale reading for the ac voltage is incorrect. The random uncertainties are correct.

6 Analysis

The candidate was awarded **1 out of 1 mark** because they have correctly calculated the gradient of their line and stated its significance.

7 Conclusion

The candidate was awarded **1 out of 1 mark** because the conclusion answers the aim.

8 Evaluation

The candidate was awarded **1 out of 3 marks** because they have provided three evaluative statements, however two of the statements cannot be awarded a mark. In statement 1, the justification is insufficient. Statement 2 is awarded a mark. The candidate has identified an issue with literature data and stated its effect. Statement 3 identifies an issue but offers no solution.

9 Structure

The candidate was awarded **1 out of 1 mark** because the report is clear and concise and has an acceptable title.

Overall

The candidate was awarded a total of **12 out of 20 marks**.

Candidate 4

1 Aim

The candidate was awarded **1 out of 1 mark** because their aim describes the purpose of the investigation.

2 Underlying physics

The candidate was awarded **2 out of 3 marks** because they have shown a reasonable understanding of the physics involved in their investigation. Relationships have been stated and the terms defined.

3 Data collection and handling

The candidate was awarded **3 out of 5 marks**. The marks were awarded as follows:

3(a) **0 out of 1 mark** was awarded because the summary states what will be measured and how it will be measured. However, there is no description of what is to be altered.

3(b) **1 out of 1 mark** was awarded because the candidate has a range of five values over a limited set of distances. This is just acceptable. The candidate has repeated measurements of the dependent variable for each value.

3(c) **1 out of 1 mark** was awarded because the candidate's table has correct units and headings. The mean values are correctly calculated and tabulated, the sample calculations do not have units, but they are given in the table.

3(d) **1 out of 1 mark** was awarded because the data from the internet source is relevant to the candidate's experiment.

3(e) **0 out of 1 mark** was awarded because the source is referenced but there is no citation.

4 Graphical presentation

The candidate was awarded **3 out of 3 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because the candidate has drawn a scatter graph with suitable scales.

4(b) **1 out of 1 mark** was awarded because the axes of the graph have suitable labels and units.

4(c) **1 out of 1 mark** was awarded because the candidate has plotted the points accurately and drawn an acceptable line of best fit.

5 Uncertainties

The candidate was awarded **2 out of 2 marks** because they have given plausible reading uncertainties. The random uncertainties have been calculated correctly.

6 Analysis

The candidate was awarded **0 out of 1 mark** because although the gradient is calculated correctly and inserted into the relationship, the further analysis that then follows is invalid as the graph does not pass through the origin.

7 Conclusion

The candidate was awarded **0 out of 1 mark** because the conclusion stated by the candidate is invalid as the line does not pass through the origin.

8 Evaluation

The candidate was awarded **1 out of 3 marks** because although they make four evaluative statements, only one of them is awarded a mark. Statement one is insufficient to award a mark. Statement two is not awarded a mark as a reduction in random uncertainty would improve the precision but not the accuracy. Statement three is correct as measuring the background light and subtracting it from the reading would help reduce the systematic uncertainty. Statement four is not awarded a mark as ruler with 1 mm divisions is already being used.

9 Structure

The candidate was awarded **1 out of 1 mark** because the report is clear and concise and has an informative title.

Overall

The candidate was awarded a total of **13 out of 20 marks**.

Candidate 5

1 Aim

The candidate was awarded **1 out of 1 mark** because this aim is acceptable.

2 Underlying physics

The candidate was awarded **1 out of 3 marks** because they demonstrate a limited understanding of the underlying physics.

3 Data collection and handling

The candidate was awarded **5 out of 5 marks**. The marks were awarded as follows:

3(a) **1 out of 1 mark** was awarded because the method contains details of what is to be measured and how the experiment was carried out. The diagram aids clarity.

3(b) **1 out of 1 mark** was awarded because the candidate has sufficient data. The range of readings is appropriate to the aim and the candidate has repeated measurements of the dependent variable for each value.

3(c) **1 out of 1 mark** was awarded because the candidate has constructed two tables which are considered together. Both tables have appropriate headings and units. All the mean and derived values are correctly calculated.

3(d) **1 out of 1 mark** was awarded because the data from the internet source is relevant to the candidate's experiment.

3(e) **1 out of 1 mark** was awarded because the source is cited beside the data and referenced later in the report. A date is included.

4 Graphical presentation

The candidate was awarded **2 out of 3 marks**. The marks were awarded as follows:

4(a) **1 out of 1 mark** was awarded because the candidate has drawn a scatter graph with suitable scales.

4(b) **1 out of 1 mark** was awarded because the axes of the graph have suitable labels and units.

4(c) **0 out of 1 mark** was awarded because the line has been forced through origin and so the mark is not awarded.

5 Uncertainties

The candidate was awarded **1 out of 2 marks** because a mark is awarded for the scale reading uncertainty stated in the section 'uncertainties', both attempts at random uncertainties are missing a unit and so the second mark is not awarded.

6 Analysis

The candidate was awarded **0 out of 1 mark** because they have correctly calculated the gradient of their line and stated its significance. However, the equation stated to calculate the gradient is incorrect and so the mark is not awarded. Refer to General Marking Principle 11, example 2.

7 Conclusion

The candidate was awarded **1 out of 1 mark** because the conclusion answers the aim.

8 Evaluation

The candidate was awarded **2 out of 3 marks** because although they make three evaluative statements, only two of them are awarded a mark. The first statement concerning the temperature is not awarded a mark as the candidate has not stated that the temperature will affect the refractive index. The second statement is awarded a mark as the candidate has identified a problem with conducting the experiment and offered a valid solution. The last statement indicates that increasing the number of repeated readings will decrease the random uncertainty due to the variation in the results.

9 Structure

The candidate was awarded **1 out of 1 mark** because the report is clear and concise and has an informative title.

Overall

The candidate was awarded a total of **14 out of 20 marks**.

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